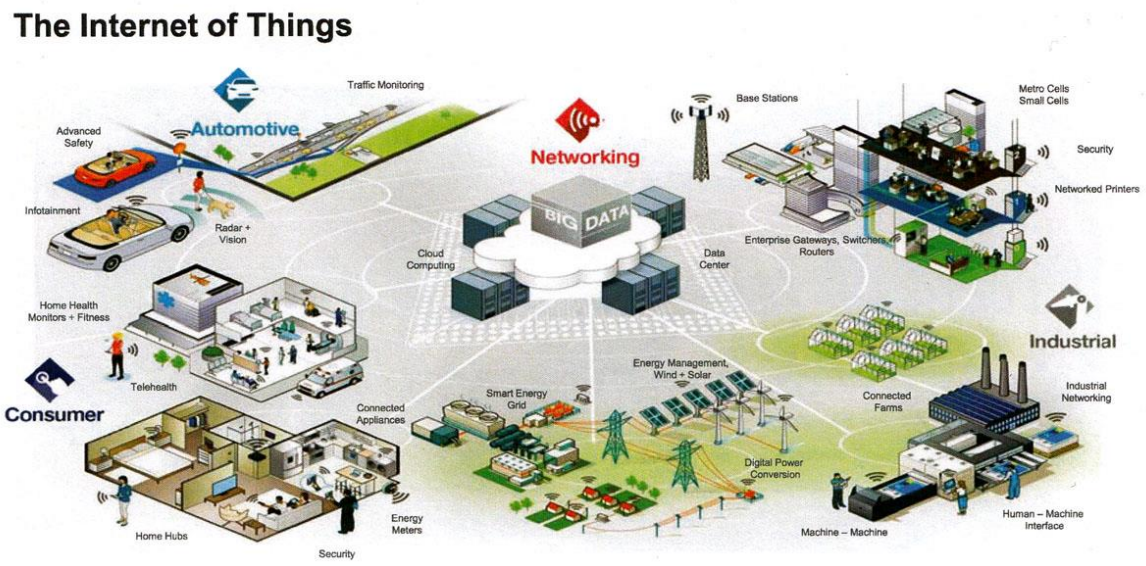
Internet of Things – IoT

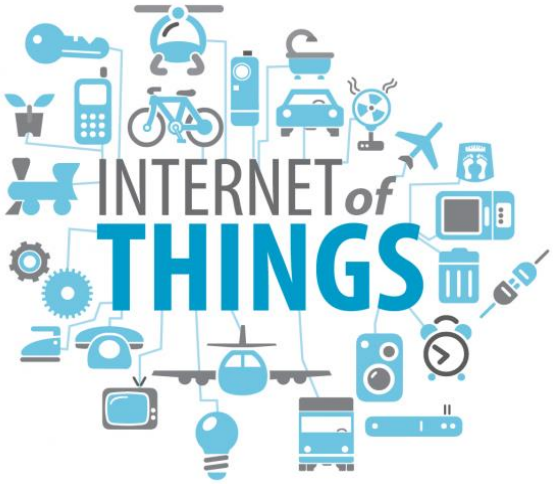
http://www.electronicshub.org/internet-of-things/
http://www.1ologic.com/technology/iot-technology.html
http://www.semi-journal.com/ทำความรู้จักกับ-mqtt-cloud/



1. Introduction

1.1. What is Internet of Things?

Internet of Things is a world where everything around us like phones, TV's, lights, refrigerators, AC's, automobiles etc. are monitoring, sensing and interacting among themselves with or without human control. Many consider Internet of Things or IoT as the next technical revolution in the internet.



Internet of Things (IoT) is a network of embedded systems (devices) with internet connectivity, which allows them to connect to and interact with other embedded devices, services (machines or appliances) and people on a large scale.

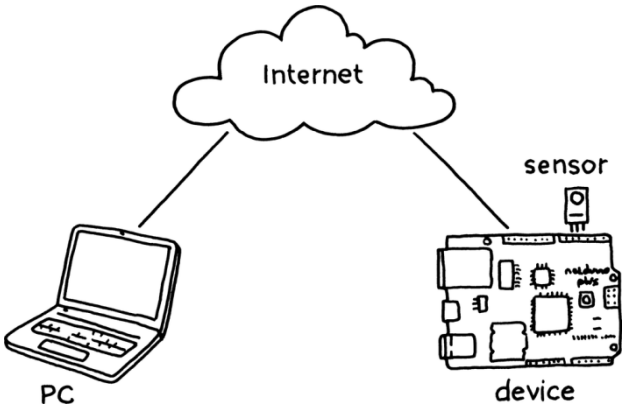
Internet of Things (IoT) is one of the most fascinating trends in controlling variety of things or objects intelligently through wired or wireless communication systems.

It enables the things to be connected or controlled anytime, anyplace with anyone or anything using any path or network and any service. The main aim of Internet of Things (IoT) is to make different tasks much easier for user control and monitoring.

With the help of Internet of Things (IoT), home or office automation systems, environmental or biological monitoring, smart grids etc. can be interconnected, allowing them to share the information between them that affect each other's performance.

IoT consists of things or devices that have unique identities and are connected to the internet through a communication network.

It refers to the network of physical objects that are embedded with electronics, sensors, actuators, software and communication connectivity, in which the whole arrangement enables the exchanging of data, remote sensing and control of various objects or things.



PC and a Device connected to internet

This concept can be viewed as connecting any device by shifting its ON and OFF switch to the Internet.

Using IoT, all objects in daily life such as washing machines, lamps, coffee makers, air conditioners, etc. are equipped with identifiers and wireless connectivity in order to provide remote control and exchange of information while executing meaningful applications towards a common user or machine goal.

Physical Objects
+
Controller, Sensors and Actuators
+
Internet
=
Internet of Things (IoT)

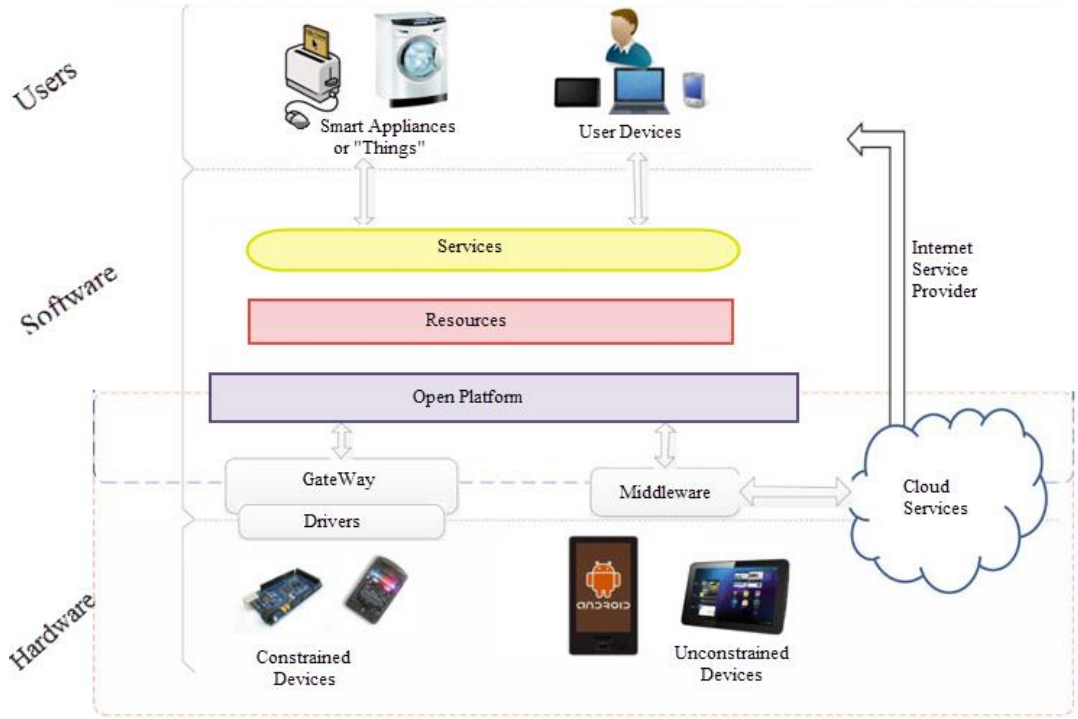
A simple equation for the Internet of Things is shown in above figure in which a physical object follows the function of the thing and being connected to the internet, it can be controlled and monitored through internet.

The sensor within or attached to the objects are connected to the internet via wired or wireless internet connections. Various local area connections for these sensors include ZigBee, Bluetooth, RFID, Wi-Fi, etc. These sensors also use wide area networks including GSM, GPRS, 3G, 4G, etc.

1.2. IoT Architecture

The concept of Internet of Things (IoT) is not entirely new as the fields of telecommunication, industrial control and process control are already using it.

But to implement the concept of Internet of Things to the latest trends, numerous technological architectures are being developed around the feasibility and applicability of Internet of Things.



A reference architecture is proposed which focuses on providing a complete solution to facilitate design, development and readiness of the smart environment as per the Internet of Things model. The following figure shows a simplified architecture of Internet of Things (IoT) domain.

As shown in the figure, the architecture of the Internet of Things model can be further subdivided into three major layers. The lowest layer consists of the hardware community, which again is divided into two groups of devices.

The first groups of devices are constrained devices which have limited resources and features and hence rely on other devices to perform some processes. The external devices are smart gateways which possess a threat to expose the functionality to the clients.

The second groups of devices are unconstrained devices which have enough features and resources that are necessary to run processes.

Even if the unconstrained devices lack the necessary feature to perform a particular process, they have middleware components that provide the functionalities directly to the client via a platform or third party cloud service.

The next layer or the middle layer in the architecture of the Internet of Things is the software layer, which supports an open source platform.

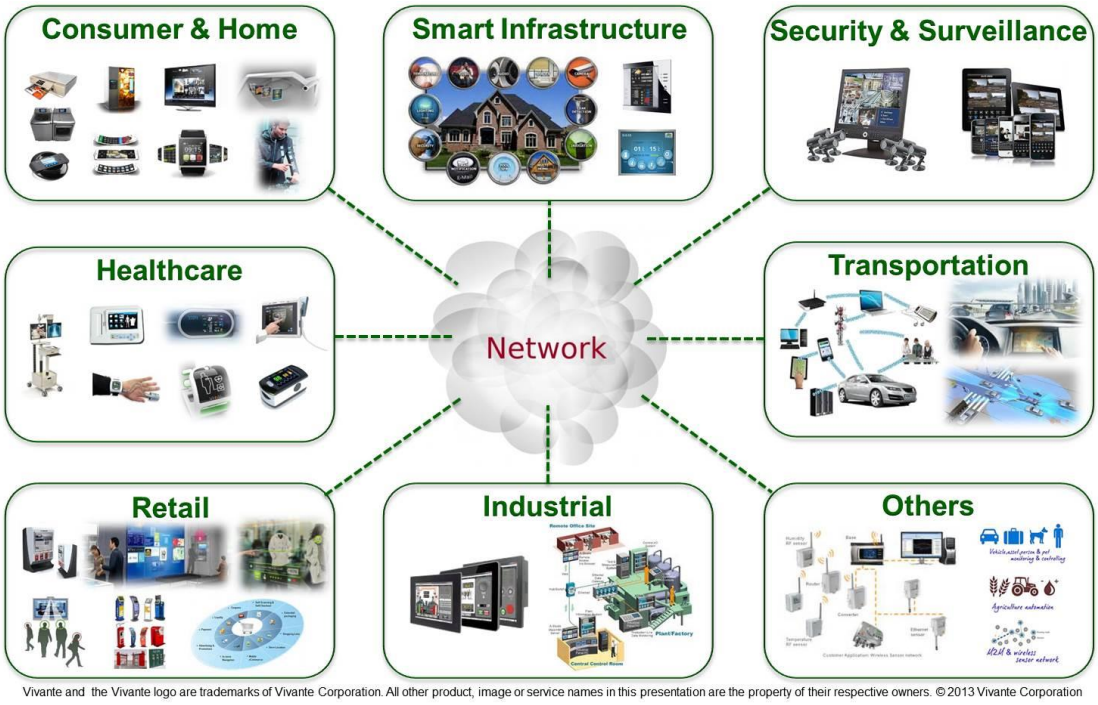
The task of this layer is to provide a mechanism to define and setup the functionalities of the hardware like sensors, actuators, process handling etc. and also organize them I order to build the services (either simple or complex).

The software level also has the task of implementing necessary protocols, connectivity drivers and communication standards.

The final layer in the architecture Internet of Things is the user layer. This layer consists of clients which make use of the services provided by the software layer. The clients can be smart phones, TV's, laptops, smart machines, home appliances etc.

1.3. Possible Applications of IoT

Different technologists, books and research studies have different views on the possible applications of Internet of Things (IoT). Some researches divide the applications of IoT into three domains. They are



- Individual – for smart living
- Industry – for efficient business operations
- Infrastructure – for smart cities

A small list of many possible applications of IoT in each of the domains is mentioned below.

1.4. IoT Applications

For individual applications, which enable smart living, the set of applications of IoT can be further divided into three categories. They are

1.4.1 Home

- Security systems, surveillance systems, alarm systems (smoke, motion, gas etc.)
- Energy monitoring systems like lighting, thermostat, home appliances, energy meters
- Water management systems like motor control, level control, sprinkler system etc.
- Home entertainment systems like audio, video, projectors, etc.
- Personal health monitoring systems like blood pressure, diabetes, ECG etc.

1.4.2 Consumer electronics

- Wearable devices like watches for quality living
- Fitness monitoring and tracking for smart living
- Kids, elders and pets monitoring for safe living
- Leisure and entertainment for quality living

1.4.3 Smart car or transport system

- Driverless or autonomous driving
- Traffic information
- Vehicle diagnostics like engine, oil, brake etc.
- Location based services like GPS

For industries to run business efficiently, an intelligent system to control and organize the process. Some of the possible markets associated with industry are mentioned below.

- Agriculture includes irrigation, production, cattle, livestock etc.
- In factories, the concept of automation, robotics, machine and process control etc.
- Construction and civil works include smart buildings and offices, heating, air conditioning, ventilation, HVAC, lighting, energy monitoring etc.
- Communication
- Manufacturing
- In health sector, health monitoring, clinical and research labs, diagnostics, treatment and insurance.
- Smart shops include vending machines, ATMs, electronic point of sale, smart retail units, hospitality etc.
- Smart environment include surveillance and monitoring of air and water for pollution.
- Smart grids like water grid, energy grid, gas and oil pipelines etc.

In order to build and develop smart cities and communities, the main ingredient is the smart infrastructure. The possible applications in infrastructure of smart cities are as follows:

- Smart education
- Security, defense and disaster management
- Public safety includes ambulance, police, fire and surveillance.
- Public transportation includes trains, buses, planes, cargo, smart vehicles, etc.
- Highways include lighting, parking, tolls, meters etc.

1.5. IoT Applications

The Internet of Things (IoT) can be used for a wide variety of applications including homes, energy systems, agriculture, health, industry, logistics and environment.

Some of the most prominent application areas of IoT are discussed below. Based on a specific application domain, IoT products mainly categorized into smart wearable, smart city, smart home, smart environment and smart enterprise.

1.5.1 Building and Home Automation

Smart lighting saves the energy by adapting the lighting to the ambient conditions and dimming or switching ON/OFF the lights when needed. This smart lighting is achieved by using solid state lights (such as LED lights) and IP-enabled lights.

By detecting the environmental changes and human movements, the lights are controlled using the smart devices. IoT applications like mobile apps and web applications enable the remote control these wireless-enabled and internet connected lights.

Smart appliances in homes make the management and control much easier than appliances having its own controls or remote controls. Internet of Things allows the user to get the status information and also remote control capability remotely.

Some of the appliances work based on IoT include smart refrigerators (that keep tracks of various items stored using RFID tags), smart thermostats to control the temperature, smart washer/dryers, smart TVs, etc.

Intrusion detection system uses various sensors (like PIR and door sensors) and security cameras in order to detect intrusions and then to give alarming. The IoT devices on homes provide the intrusion alerts in the form of an SMS or an email to the user and nearby police.

The intrusion systems are based on Universal-Plug and-Play technology, which generates the intrusion messages by using image processing techniques to recognize and extract intrusion subject.

Smoke and gas detectors in homes and buildings detect smoke (which is the early sign of fire) and harmful gases (such as carbon monoxide, LPG, etc.).These smart sensors can raise alerts and sends an email or SMS to the user or local safety department.

1.5.2 Cities

Smart traffic and parking systems use IoT based sensor and circuitry for transferring the data over internet. Sensor detects the number of empty parking slots and correspondingly sends the information to the main database over internet.

Then smart parking application in smart phones, tablets and in-car navigation system continuously displays the parking information to the driver.

Similarly, traffic jams on roads can be reduced by employing distributed system of sensor networks which can capable of sensing information on roads and to convey the information to main server over internet.

Smart lighting saves the energy by controlling lights on roads, parks, buildings, etc. These smart lights are connected to the internet to achieve the remote control, lighting schedules and lighting intensity control.

The sensors attached with the lights are capable of controlling the lighting depends on the ambient conditions and also communicates with other lights (circuitry) to exchange the information.

Smart surveillance system ensures the safety and security by monitoring infrastructure, public transport and events. This system consists of a large number of internet connected and distributed video surveillance cameras which sends the information to central cloud-based storage system.

1.5.3 Environment

IoT based weather monitoring system collects the various sensors data (temperature, pressure, humidity, etc.) and sends to the cloud-based applications.

The data collected by the cloud, further analyzed and visualized on IoT based weather monitoring applications on mobiles, computers and other displaying devices. It also sends the weather alerts to the subscribed users from the cloud-based applications.

IoT based air and noise pollution monitoring systems can monitor the harmful gases in the air caused by automobiles and factories and also noise levels in the given environment by using various sensors.

Gaseous and metrological sensors are used for monitoring the air pollution and are deployed at several distributed monitoring station. All these stations are interconnected to one another using machine to machine communication.

Similarly, in case of noise monitoring systems various distributed noise monitoring stations are deployed at various places, from them noise data is aggregated in the servers or in the cloud.

IoT based forest fire detection systems uses a number of sensory or monitoring nodes which are deployed at various places in the forest. This system achieves the early detection of fires by detecting the various sensors data (temperature, light levels, humidity, etc.) at nodes.

1.5.4 Energy

Smart grids collect and analyze the data collected from various electrical grids and correspondingly provide predictive information and recommendations to the utility companies. The monitoring of health of an equipment and integrity of grid can be evaluated by using IoT based sensing and measuring technologies.

Smart meters can capture real time energy consumption from various customers and which are capable of transferring data remotely to main server, remotely switching ON/OFF the power supply when needed and prevents the power thefts.

IoT based systems with transformer at the point of interconnection of various renewable energy sources to grid, helps to determine various electrical variables and hence grid stability and reliability problems are avoided.

1.5.5 Agriculture

Smart irrigation systems save the water and power while improving the crop yields. Soil moisture sensors along with IoT devices determine the amount of soil moisture and accordingly turn the irrigation pumps.

Also, this smart irrigation system collects the soil moisture data continuously on the server, which can be utilized for planning watering schedules.

1.5.6 Industries

IoTs plays an important role in industrial applications for monitoring the machine operation conditions, prognosis and diagnosis of machines during faults and remote controlling the machines when needed.

IoT devices with sensor and actuators provide the real-time optimization of manufacturing supply chains networks and its production.

1.5.7 Medical and Healthcare Systems

Wearable IoT devices allow continuous monitoring various physiological parameters that provides emergency health notification system and remote health monitoring systems.

These wearable devices with sensors like body temperature, pulse rate, heart rate, blood pressure, etc. collects the health parameters data, determines the abnormalities and correspondingly generates the alarm or notifications to the users or doctors.

1.5.8 Logistics

IoT devices can drive the various transportation systems effectively by implementing interconnection between vehicles and drivers or users.

It can provide advanced route guidance, vehicle routing and scheduling by combining route patterns and transportation schedules based on the availability of vehicles. The smart transportation system includes vehicle tracking, automatic toll collection, and safety and road assistance systems.